

Mohana Nitej Mithinti

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PROFESSIONAL SUMMARY

GenAI / Agentic AI Engineer with 4+ years of software engineering experience shipping production-grade LLM applications across energy, manufacturing, HSE, and financial workflows. Hands-on with multi-agent orchestration (LangChain, LangGraph), Agentic RAG, NLQ-to-SQL agents, layered prompt engineering, and async FastAPI backends on Azure OpenAI, Azure AI Foundry, and Databricks Mosaic AI. Proven track record leading POCs end-to-end — architecture through client demos and Azure deployment — while mentoring engineers and partnering with cloud, data, and security teams.

TECHNICAL SKILLS

- **Languages:** Python, TypeScript, JavaScript, SQL, PL/pgSQL, Java, C/C++, R
- **GenAI & Agents:** LangChain, LangGraph, CrewAI, LlamaIndex, MCP, OpenAI Function Calling, Hugging Face Transformers; Azure OpenAI (GPT-4, GPT-4o, GPT-4 Vision), Azure AI Foundry, Azure Speech Services, Azure Document Intelligence, AWS Bedrock
- **RAG & Vector Stores:** Azure Cosmos DB (MongoDB vCore vector index), pgvector, Pinecone, Chroma, Azure Cognitive Search, ElasticSearch; semantic chunking, embedding pipelines, layered prompt engineering
- **Backend & Data:** FastAPI, async/await, REST, JWT auth, semaphore-gated concurrency, background tasks; PostgreSQL, MongoDB / Cosmos DB, MSSQL, Oracle, Microsoft Fabric (Lakehouse, OneLake), Databricks Mosaic AI
- **Cloud, DevOps & MLOps:** Azure Container Apps, App Service, Key Vault, Azure DevOps; AWS (Lambda, SageMaker, ECR); Docker, Kubernetes, GitHub Actions, Jenkins, MLflow, LangSmith, CI/CD
- **Frontend, BI & ML:** React, Tailwind, Chart.js, Power BI Embedded, Tableau, ReportLab, matplotlib; Pandas, NumPy, PyTorch, Apache Spark, Hive, Hadoop, Kafka

PROFESSIONAL EXPERIENCE

Software Engineer — GenAI — Biz-metric Partners Inc., Houston, TX

Feb 2025 – Present

- Architected and shipped a multi-tenant FastAPI GenAI platform (JWT auth, Azure Container Apps, AES-128 encrypted secrets) serving engineers across subsea, manufacturing, and HSE customers; supports unstructured RAG and NLQ-to-SQL across isolated client schemas.
- Built dual-mode conversational AI: a LangGraph state machine for Agentic RAG over Azure Cosmos DB (1536-dim vectors, cosine similarity, 20K+ chunks) plus a LangChain SQL Agent translating NLQ into validated PostgreSQL queries — improving answer relevance 35–45% and reducing manual analyst effort ~60%.
- Designed a 4-layer modular prompt architecture (foundation → schema → query templates → runtime business rules) with auto-routing across module dependencies; cut hallucinations ~40% via groundedness checks, citation enforcement, and read-only SQL validation.
- Engineered a 7-tool multi-agent troubleshooting system in LangGraph (data retrieval, statistical compliance, regression-based RCA, RAG over SOPs) for a USG manufacturing client — integrating on-prem MSSQL and Oracle with metadata-driven actionability ranking.
- Shipped parallel LLM pipelines (up to 10 concurrent Azure OpenAI calls, semaphore-gated) for an HSE analytics platform auto-generating executive summaries, intervention detection, and PDF/email reports — replacing manual analyst workflows.
- Architected a CITGO Split Billing Compliance chatbot on Microsoft Fabric Lakehouse with OneLake Shortcuts, Power BI Embedded, and Azure Speech Services — eliminating sync pipelines and enabling voice-driven NLQ.
- Hardened production with two-tier query caching, exponential-backoff retries, rate-limit-aware embedding batching (240K TPM), PG connection pooling, and LangSmith-based LLMops — cutting response latency ~30%; drove POCs on MCP, Azure AI Foundry, Databricks Mosaic AI, and a hybrid CV + LLM pipeline for P&ID / HAZOP automation; mentored juniors and led architecture reviews for scalability, security, and enterprise compliance.

AI Full Stack Developer — Arizona State University, Tempe, AZ

May 2024 – Jan 2025

- Cut AWS Lambda compute time 95% via semantic chunking and batched LLM embedding requests on a RAG pipeline indexing Google Drive content into OpenSearch.
- Containerized the application and shipped CI/CD: SageMaker testing → ECR → AWS Lambda.

Software Engineer — Algonox Technologies Pvt Ltd, Hyderabad, India

Aug 2020 – Jun 2022

- Designed and maintained a Python FastAPI microservice on Azure (SQL DBs, Storage, App Services); led REST services, CI/CD, Docker, EC2 — cutting maintenance cost 70% and lifting business-requirement throughput 25%.
- Resolved 700+ complex technical cases (90% quality index), authored 30+ KB articles, automated ETL scripts (case lifecycle ↓ 50%), and mentored 3 interns; engaged with clients including L'Oréal and Capgemini.

EDUCATION

Arizona State University, M.S. Computer Science (GPA 3.97) — Tempe, AZ

Aug 2022 – May 2024

KL University, B.Tech. Computer Science Engineering (GPA 3.25) — Guntur, India

Jun 2016 – Jun 2020

Certifications & Awards: OCI 2024 Generative AI Certified Professional · DP-700 Microsoft Fabric Data Engineer Associate · AZ-104 Microsoft Azure Administrator · CBAP · Sparkling Star Award (Bizmetric).